Build a complete prediction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect net force, velocity change, and final position in one model.

SUCCESS CRITERIA - I can...

1. Connect force, acceleration and position consistently.
2. Show quantities and units at every stage.
3. Explain the assumptions and limits of the prediction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “reference frame” and “net force” to explain your model or result. Sentence starters are optional.

reference frame: the origin and directions used to describe motion

net force: the vector sum of all forces on one object

prediction: what a model says may happen next

01 Bring back an earlier idea

A 3 kg box has μ_s=0.50 and μ_k=0.40 on a level floor; g=9.8.
Compare pushes of 6 N and 18 N, starting from rest.
Find actual friction and acceleration.

Word help: friction = a contact force opposing relative sliding or its tendency

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $a = \Sigma F/m$; $v_f = v_i + a\Delta t$; $x_f = x_i + \frac{1}{2}(v_i + v_f)\Delta t$. Constant acceleration; use one frame. i = initial; f = final. Apply each direction separately.

Worked reasoning: A 2 kg cart starts from rest under 6 N right and 2 N left for 2 s. Net force 4 N gives $a=2\ m/s^2$. Its displacement is $\frac{1}{2}\times2\times2^2=4$ m right.

Your turn: Draw the force diagram and the matching velocity-time triangle.
Check the 4 m using its area.

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 End point versus displacement

Fictional probe: x_i=-100 km, y_i=0; v=(4, 3) km/s for 20 s constant.
Find end point, displacement magnitude, and distance from origin.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.



The small starting dot is the object; the labeled circle marks Earth at the origin. Not to scale.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Add bounds

For 1, initial x bound±1 km; each velocity component has independent bound±.1 km/s; y_i and time exact.
Find end point coordinate intervals.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

05 A new force

A 1000 kg probe experiences an additional constant 200 N net force along+x for 10 s.
Find the extraΔv_x and extra displacement relative to constant velocity with the same initial state.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

06 Choose a method

A car starts at +35 m/s and brakes at -5 m/s² until rest. A student extends that acceleration to 9 s.
Find the stop time/distance and explain why extension is invalid for ordinary braking.

Word help: acceleration = rate of change of velocity

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

07 Exit check

Write a short public briefing for 1-2. Include a predicted position, a measurement range, and two limitations that prevent a real asteroid collision forecast.

Sentence starter: I predict ... . My model supports this because ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A cart starts at x=3 m with v=-2 m/s. Its mass is 2 kg and constant net force is +6 N for 4 s.
Predict final velocity and position.
Check whether it turns.

Sentence starter: I predict ... . My model supports this because ... .

Marzano check - circle: 1 2 3